

ISAAC SBAI METAICH

ROBOTICS, ELECTRONICS AND MECHATRONICS ENGINEER

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WORK AND RESEARCH EXPERIENCE

Academic & Research Experience

Aalborg University – Student Assistant 2025 – 2026
Aalborg, Denmark
Designed and implemented a socially-aware navigation framework in ROS 2 integrating A* global path planning with Artificial Potential Fields for dynamic obstacle avoidance. Developed embedded low-level firmware on STM32 microcontrollers to interface, calibrate, and actuate a custom flexible gripper mounted on an industrial Universal Robots (UR) manipulator.

Industry Experience

Macco Robotics – Robotics Engineering Intern 2024
Málaga, Spain
Contributed to the end-to-end development, mechanical assembly, and commissioning of an automated commercial coffee bean dispensing machine built from scratch for supermarket deployment. Designed and ran systematic multi-cycle stress tests to evaluate mechanical tolerances and isolate electrical and firmware bugs. Collaborated on programming and motion control for humanoid service robots.

Maker & Prototyping Experience

Skapalab MakerSpace – Prototyping Participant 2022 – 2024
Málaga, Spain
Engineered a high-temperature 3D printer from scratch in Autodesk Fusion designed specifically for high-performance engineering polymers (PEEK and PSU). Designed and manufactured custom mechanical input devices, managing the complete pipeline including schematic capture, PCB layout, manual component soldering, and microcontroller firmware.

EDUCATIONAL BACKGROUND

MSc. in Robotics – Aalborg University 2024 – 2026
Aalborg, Denmark
THESIS: Researched and developed a group engagement detection algorithm for humanoid robots to assess human attention and interaction cues in real time, validating tracking accuracy and robustness across simulation environments and physical robotic platforms.

- **Downward-Facing Visual Odometry:** Built a complete monocular visual odometry pipeline using Shi-Tomasi corner detection, Lucas-Kanade optical flow, and RANSAC outlier filtering for robust ego-motion estimation under sensor noise.
- **Brain-Computer Interface (BCI):** Developed a ROS-integrated BCI framework to control a Kinova Gen2 robotic arm using classified EEG signals, implementing feature extraction and decoding models.

BSc. in Electronic, Robotics and Mechatronics Engineering – University of Málaga 2020 – 2024
Málaga, Spain
Acquired solid foundations across electronics, control theory, robotics, and mechanical systems, specializing in industrial automation, PLC programming, sensor integration, and embedded systems development.

Machine Learning and Visual Data Analysis Course 2023 – 2024
Spain
Practical specialization covering supervised and unsupervised learning, classification algorithms, regression modeling, data preprocessing, and statistical model evaluation techniques.

SOFTWARE & HARDWARE SKILLS

Programming: Python (PyTorch, OpenCV), C, C++, MATLAB, VHDL, Rapid
Robotics & Middleware: ROS 2, ROS, Gazebo, Kinematics, Path Planning
Hardware & CAD: Autodesk Fusion 360, AutoCAD, PCB Design, Soldering, STM32, Arduino
Automation & Tools: TIA Portal, TwinCAT, Linux, Git, Docker

LANGUAGES

Spanish:	Native
Moroccan Arabic:	Spoken
English:	C1 Fluent
French:	B1 Intermediate